



K. K. Wagh Institute of Engineering Education & Research

Department of Artificial Intelligence and Data Science

A.Y:2025-2026

Project Work

PID:04

GrazeScale:Animal Weight Estimation System

Abstract

The increasing demand for efficient livestock management has created a need for accurate and non-invasive methods to estimate the weight of cattle. This project presents the development of an automated weight estimation system based on image analysis of cattle captured through smartphone cameras. By acquiring images from multiple viewpoints, including side and back views, the system extracts key physical features such as oblique length, girth, body length, and width, which are critical indicators of the animal's size and weight. These measurements are then used as input to a predictive model that estimates the weight of the cattle. The proposed method eliminates the need for physical weighing scales, reducing stress on animals and labor costs for farmers. The system aims to provide a practical, cost-effective, and accurate solution for cattle weight estimation that can be easily deployed in field conditions, thereby supporting improved decision-making and productivity in livestock management.

Keywords: *Weight Estimation, Cattle Weight Prediction, Image Processing, Deep Learning, YOLOv8, Smartphone Camera, Livestock Management, Object Detection, Regression Model, Non-invasive Measurement*

Group Members	Division-Roll Number	Name Of Project Guide
Akanksha V. Parkhe	A-13	Prof.W.W.Pingle
Lulwa S. Anif	A-14	
Pranjal D. Tile	A-15	
Sakshi S. Joshi	A-16	